IntradayBot — Trading Strategy Document

System: Automated Nifty 500 Intraday Momentum Bot **Stack:** Python / FastAPI / MySQL / Upstox API / Claude (claude-sonnet-5) **Style:** Momentum breakout/breakdown trading with a hybrid quant + AI decision gate

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1. Strategy Thesis

The bot trades **intraday momentum continuation** setups on liquid Nifty 500 names. The core idea:

1. Find stocks that are *already* showing unusual strength (volume + trend) on the daily chart.
2. Narrow that list continuously to whichever names are *actively* moving right now.
3. Use rigid, rule-based 5-minute triggers to catch the moment momentum accelerates.
4. Mathematically pre-compute Entry/Stop-Loss/Target so risk is bounded *before* any subjective judgment is applied.
5. Use Claude only at the very end, as a visual/structural sanity check — not as a source of price levels.

This "quant-first, AI-last" ordering is deliberate: Claude cannot invent numbers, cannot skip risk controls, and only ever narrows the funnel (ENTER/SKIP), never widens it.

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2. The Five-Layer Pipeline

Layer 1 — Morning Pre-Filter (9:00 AM, daily)

Reduces the ~500-stock universe to a focused watchlist (typically 80–120 names) using **daily** candles (45-day lookback). A stock must pass **all** of:

Rule	Logic	Threshold
Volume Activity	Yesterday's volume ÷ 20-day avg daily volume	⩾ 1.3x
Trend Filter	Close vs 20-day EMA	Close > EMA20 (LONG) / < EMA20 (SHORT)
Breakout Proximity	Distance from 20-day high/low	⩽ 8% away (not extended)
Relative Strength	Stock's 5-day return vs Nifty 50's 5-day return	Must beat/lag Nifty in the trade's direction

A fail-safe external scraper can append additional "breakout" stocks discovered outside the mathematical filter, but only if the pipeline's own filters fail gracefully — it never overrides the core rules.

Layer 2 — Active Watchlist Manager (hourly + 10-min drift check)

Keeps the live-tracked pool lean to conserve WebSocket bandwidth:

- **Drops** stocks that are simultaneously: RSI(14) between 44–56 (sideways), 3-candle volume below the 20-bar average, and a flat 10-candle price slope.
- **Adds** stocks from the morning list (or freshly scraped breakouts) that show a sudden 3-candle volume surge (⩾ 2x the 20-bar average).

Layer 3 — Quantitative Signal Scanner (every 5 minutes, 9:20 AM–2:30 PM)

Pure rule-based, no external API calls. For each active-watchlist stock, evaluates 5 conditions on the most recently **completed** 5-minute candle:

1. **Volume Expansion** — candle volume ⩾ 2x the 6-bar average
2. **VWAP Reclaim** — price > +0.15% above VWAP (LONG) / < −0.15% below (SHORT)
3. **Momentum (RSI-14)** — 58–75 (LONG) / 25–42 (SHORT)
4. **Micro-Trend Alignment** — 9 EMA vs 21 EMA in the trade's direction
5. **Candle Strength** — body/range ratio ⩾ 0.50 (closes near the high/low, minimal wick)

Dynamic threshold: normally 3/5 conditions required; tightens to **4/5** during lunch hour (12:00–1:30 PM) or on a classified LOW_PARTICIPATION day.

Additional filters layered on top:

- *Climax volume skip* (LONG only): if candle volume ⩾ 5x the 20-bar average, skip — treated as exhaustion, not fresh momentum.
- *15-minute multi-timeframe alignment*: 15-min EMA20 must be sloping in the trade direction and price must be on the correct side of the 15-min VWAP, or the signal is discarded.
- *Sector momentum filter*: LONGs are skipped if the stock's sector index is down >0.3% on the day (and vice versa for SHORTs).

Output: top 5 signals ranked by conditions met, then volume ratio.

Layer 4 — Mathematical Target & Risk Engine

For each qualifying signal, deterministically computes the trade geometry **before** Claude ever sees it, so Claude cannot alter risk parameters.

- **Adaptive R:R multiplier** based on live market regime: 2.5x (trending), 1.5x (ranging/high-vol), 2.0x (default).

- **Entry** = LTP ± 0.05% slippage buffer.
- **Stop-Loss** = 20-bar structural support/resistance ± a small buffer.
  - If that distance is *too tight* (< 0.3% of price) → widen to 1x ATR(14).
  - If that distance is *too wide* (> 1.5x ATR(14)) → **reject the trade outright** (hard risk cap).
- **Target** = Entry ± (multiplier × risk), but **rejected** if it would land beyond the 20-bar resistance/support — unless volume is confirming a genuine breakout (6-bar vol ratio > 3.0 and volume trend expanding), in which case the target is capped just past the level (0.5%).

This layer also assembles a rich feature payload for Claude: price vs VWAP/EMA20/20-day-high, RSI (5-min and daily), EMA slope, candle strength, volume trend/divergence, support/resistance distances, previous-day high/low context, sector performance, Nifty/Bank Nifty trend, India VIX, and the last 20 completed 5-minute candles (OHLCV) as the visual "chart" Claude reasons over.

Layer 5 — AI Setup Validation (Claude)

Claude's role is intentionally narrow: **it validates structure, it does not calculate anything.**

It receives the Layer 4 payload and returns a JSON decision:

- **Confidence 1–2** → automatic SKIP
- **Confidence 3** → ENTER only if zero red flags
- **Confidence 4–5** → ENTER

What it's checking for: trend consistency (higher-highs/higher-lows or the inverse), a compression/consolidation pattern preceding the trigger candle, and red flags a pure-math filter might miss (e.g., a large upper wick on the trigger candle suggesting rejection).

If Claude approves, Python maps the *already-computed* Layer 4 Entry/SL/Target onto the order — Claude's output never touches price levels.

There is also a **local rule-based fallback** (`run_local_rule_analysis`) that reproduces Claude's decision logic in pure Python (VWAP direction, proximity to S/R, ATR risk check, volume-contraction check, confidence scoring incorporating Nifty trend, relative strength, daily RSI extremity, Bank Nifty alignment, and previous-day high/low clearance). This is used when the AI filter is toggled off, and is also run in shadow mode on every AI-approved trade to log whether the rules alone would have entered — enabling a clean "Claude vs. no-Claude" performance comparison.

Parallel Strategy — Opening Range Breakout (ORB)

Alongside the 5-layer momentum pipeline, a separate ORB sub-strategy observes the 9:15–9:30 AM opening range for each active-watchlist stock, then scans for breakouts of that range from 9:30–11:00 AM (1-minute cadence). ORB signals are built into an equivalent feature payload and pushed through the same Layer 5 (Claude or local-rule) validation gate

before execution, and are blocked if they contradict the day's classified Nifty regime (e.g., no ORB LONGs on a bearish day).

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3. Market Regime & Volatility Adaptation

- **Regime classification** (re-evaluated at 9:30 AM and hourly thereafter): TRENDING_UP / TRENDING_DOWN / HIGH_VOLATILITY / RANGING, derived from Nifty's daily change %, its 9:15–9:30 AM opening range behavior, and India VIX. Feeds directly into the Layer 4 target multiplier.
- **Participation ratio**: actual vs. expected 15-minute opening volume classifies the day as NORMAL / LOW_PARTICIPATION / HIGH_PARTICIPATION, which tightens the Layer 3 threshold to 4/5 on thin days.
- **VIX-based position sizing**: quantity is scaled down as VIX rises — 100% below 13, 80% at 13–16, 60% at 16–20, 40% at 20–25, and trading is **blocked entirely** at VIX ⩾ 25.
- **VIX-based caution alerts**: Telegram warnings fire when VIX crosses 18 (cautious) and 25 (halted).

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4. Position Sizing & Execution

- **Risk-based sizing**: `quantity = floor((DAILY_LOSS_LIMIT / MAX_TRADES_PER_DAY) / |Entry - SL|)` — i.e., each trade risks an equal slice of the daily loss budget.
- **Leverage**: 5x intraday (Upstox CO/MIS), validated per-symbol before every order (`check_intraday_leverage`) — trade is skipped if 5x isn't actually available.
- **Slippage guard**: order is skipped if live price has moved beyond a strategy-specific tolerance from Claude's entry (0.3% for Momentum, 0.5% for ORB).
- **Partial-fill safety**: SL is placed strictly on `filled_quantity`, not the requested quantity, to avoid an accidental naked/reverse position.
- **Broker-rejection fallback**: if the live order is rejected (e.g., insufficient margin) and Hybrid Virtual Mode is enabled, the trade continues as a "virtual" position — tracked and alerted, but not exposing real capital — rather than silently disappearing.
- **Sector concentration cap**: no more than 2 concurrent trades in the same sector.
- **Max open trades** and **max trades/day** are both hard, per-user configurable caps.

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5. Exit Management

- **Stop-Loss** is written to the database (`sl_order_id`) immediately so it survives a server crash/restart.

- **Trailing stop**: once profit reaches 1R, the bot begins trailing (adaptive ATR-based trail: looser multiple in strong trends, tighter in weak ones).
- **Time-based exits**: no new entries after 2:30 PM; all open positions are force-flattened via market order at 3:15 PM; WebSocket feed is deliberately disconnected at 3:25 PM.

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6. Risk Controls & Kill Switches

Control	Behavior
Max trades/day	Hard DB-enforced cap (default 5)
Max open trades	Hard cap on concurrent positions (default 2)
Daily loss limit	Kill-switch trips and flattens all positions if breached
Global halt flag	Admin-level switch; blocks *all* users' new entries instantly while existing positions continue to be monitored
Stale-data guard	Any tick older than 30 seconds disqualifies that stock from being scanned
Sector concentration	Max 2 trades per sector at once
Candle-completeness guard	Refuses to evaluate a 5-min candle that hasn't fully closed yet
Symbol blacklist	A symbol that's already been traded that day is not retraded

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7. Why Claude Sits Where It Does

The design puts Claude *after* every hard quantitative filter and risk check, not before:

- It cannot lower the bar Layer 3/4 already set — it can only say SKIP to something that already passed every mathematical rule.
- It never generates a stop-loss or target — those come from deterministic structural math, closing off an entire class of failure modes (e.g., "the AI hallucinated a price level").
- Its unique value-add is exactly the thing rules struggle with: reading the *shape* of the last 20 candles for consolidation-before-breakout patterns and wick-based rejection signals that a 5-condition scorecard can't see.
- A parallel, pure-Python rule engine mirrors its logic, both as a live fallback (if the AI filter is disabled) and as a shadow benchmark (`would_have_entered_without_claude`) so the system continuously measures whether the AI layer is actually adding edge versus the deterministic rules alone.

8. Suggested Areas to Track / Validate Over Time

- **Claude vs. local-rules win rate** (`/api/performance/report` already computes this) — confirms whether the AI gate is worth its latency/cost.
- **Strategy split (Momentum vs. ORB)** win rate and average R:R.
- **Time-of-day bucket** performance (Morning / Midday / Afternoon) — lunch-hour stricter threshold should be validated against realized results.
- **VIX-bucketed** win rate — confirms the position-sizing curve is calibrated correctly rather than arbitrary.
- **Regime-conditioned** performance — whether the adaptive target multiplier (1.5x/2.0x/2.5x) is actually producing better R:R outcomes per regime, or just changing win rate/frequency trade-offs.